Correction Log for Self-Report and the Underdetermination of Mechanism

Nov 3, 2026

This log records the corrections made to the essay from the first draft to the current 18-page version, grouped by what prompted them. The essay argues that language model reports of internal states underdetermine the mechanism that produces them. It draws on Lindsey (2025), Singh, Linzen, and Ravfogel (2026), Macar et al. (2026), and Lederman and Mahowald (2026).

Corrections came from two places: checking the sources, including the full text of Singh et al., and successive rounds of review comments pasted into the conversation. The four sections below cover source checks, main-text changes by section, appendix changes, and review suggestions that were declined. A last section lists what remains unverified.

Corrections from checking the sources

Reading the full text of Singh et al. (version 2, COLM 2026) and searching for Macar et al., Lederman and Mahowald, and Lindsey changed the essay in eleven places.

Topic	First draft	Correction
Singh et al. on behavioral evidence	Said they argue behavioral evidence alone is inherently insufficient, taken from the first version's abstract	The final version says task performance alone cannot satisfy the second-order condition; it needs designs with divergent predictions and, ultimately, converging mechanistic evidence. The essay now says its standard is compatible with that
Source specificity	Placed as a middle rung of a ladder of inferences	The paper's fourth footnote says specificity is neither necessary nor sufficient for second-order representation. The essay now treats it as an orthogonal diagnostic

Topic	First draft	Correction
Second-order criteria	A vague operational reading	Replaced with the paper's appendix: generalization appropriate to content, downstream use as information about processing, and content distinguishable from the cue; the norm-threshold example; and Ramsey's job description challenge
Figures from the press article	Taken from the article	Checked against Figure 3: Llama-3.1-70B 80.5% steered and 87.5% gaslight; Qwen3-32B 92.9% gaslight; three-way Llama 41.6% and 27.5%. All matched. Control false-positive rates (23.5% and 18.5%) and the Gemma-3-27B and Qwen-2.5-72B results were added
Paper's text versus its heatmap	Not noticed	The text says very few false positives in the two-way setting; the heatmap shows 23.5% and 18.5%. The essay reports the heatmap numbers
Gaslight and internal state	Said nothing internal had changed on gaslight trials	Wrong, since a changed prompt changes the computation. The finding is that a similar report arises by a different causal route, and the essay now says so
Macar et al.	Said their circuit locates the mechanism within the anomaly reading	Now says it identifies an anomaly-detection component compatible with both a direct reporting pathway and a broader monitoring architecture. Their zero false-positive result is described as behavioral, with no commitment about mechanism

Topic	First draft	Correction
Lederman and Mahowald	Treated as fitting the anomaly reading	Now notes they present content-agnostic detection as compatible with leading theories of introspection. Their result supports the detection versus identification distinction only
Lindsey dates	One date for the paper	Bibliography now gives the October 2025 publication and the January 2026 arXiv posting separately
Press summary	Not addressed	Flagged: the claim that models were merely reacting to anything unusual is stronger than the data support
Bibliography	Ji-An et al., Steinmetz Yalon et al., and Ramsey cited in prose only	Entries added from the reference list of Singh et al.

Main-text corrections by section

Each row is a change to the body of the essay, with the reason.

Section	Correction	Reason
Abstract	Opening claim narrowed to: a change in report after an intervention establishes causal dependence but does not alone identify the mediating mechanism. Consciousness and welfare boundary moved to the opening. Mention of the appendices added	An earlier wording covered even mechanistic interventions that could discriminate; the boundary read as an afterthought

Section	Correction	Reason
2. Three readings	Removed the phrase about the output distribution reading itself off; now says the report can arise through ordinary response generation without an additional representation of the altered computation	The old phrase attributed a reading operation to the non-monitoring mechanism
3. Causal dependence	Changed "predicted equally by" to "compatible with all three candidate mechanisms"; added the Steinmetz Yalon point that intervening on a representation shows causal efficacy, not introspective access; replaced a glib sentence about circuits	Compatibility is the claim, not equal quantitative predictions
4. Gaslight condition	Opening split into paragraphs; gaslight defined as a false account of the model's own state; two-way confound restated as the design not independently varying intervention status and computational atypicality; limits stated before the data; prediction sentence ends its paragraph; corrected claim about internal state; route unpacked as differing at the input stage but ending in an altered state	The earlier confound wording assumed the anomaly account was true; the paragraph carried too much

Section	Correction	Reason
4. Interpretation of data	Added that the three-way pattern is consistent with an anomaly reading but not diagnostic of it, since instruction following, option bias, mixed mechanisms, and provenance-indifferent monitoring remain compatible	Avoids treating an available alternative as the identified mechanism
5. Label prediction	Layer-0 wording changed to "without access to the contextualized hidden states from which the target labels were derived"; stated that the probe shows a possible shortcut, not a demonstrated one; two control types named (construct validity and information locality)	Layer-0 representations are themselves activations; decodability is not use
6. Evidence	Five kinds of evidence ranked by weight; matched-anomaly design stated as substantially more informative when combined with downstream-use evidence; predictive surprise called an imperfect matching measure; sentence added that the diagnostic targets provenance-representing monitors specifically	Double hedge nearly cancelled the discriminator
6. Dissociation design	Replaced a categorical sentence about holding the representation fixed with a two-part design, then stated that identifying the pathway independently of the report is itself in question; noted the circularity risk in using Macar's features	The earlier sentence was too categorical

Section	Correction	Reason
6. Relabeling control	Removed the claim that relabeling leaves only the internal route or makes input-only predictors chance by construction; added a baseline's access and resource budget, and informational exclusion versus computational difficulty	Relabeling disrupts a semantic shortcut; it does not exclude every input-based route
6. Second-order criterion	Ramsey material expanded; added the requirement that downstream computation treat the signal as information about reliability, provenance, or condition of another process; representation said to be a matter of degree; probe sentence changed to decodability versus causal role	First-order anomaly signals can also show downstream use and generalization
6. Observability	"Fully observable" changed to available for direct recording and intervention in accessible models	Access is not transparency, and proprietary models are not accessible
7. Limits	Cut to two limits (models and training; scope of the original claims) plus a theoretical note	The first limit was already argued in Section 4
8. Standard	Three questions described as distinct dimensions, not a ladder; privileged access restated as information absent from the input, not merely hard to recover; repeated safeguards trimmed	A second-order representation could concern a deterministic state; a steering effect could meet exclusion without monitoring
Citations	Numerical citations changed to author-year with natbib	Author names appear in the prose

Corrections to the mathematical appendices

The appendices were added late and then corrected in two review rounds. The first version contained claims stronger than its proofs supported.

Appendix	First version	Correction
A. Bayes corollary	Titled "No update"; said equal function classes imply data bear on hypotheses only through prior volume	Incorrect, since priors over the same functions can differ and data can favor one. Replaced by a statement that if the sets of achievable observable distributions coincide, no decision rule based on those observations alone is uniformly reliable in distinguishing the classes, with a remark that Bayesian comparison can still differ through priors
A. Parameter counting	Said fit of this kind is not evidence	Now says the availability of a fit does not by itself discriminate mechanism classes; whether a fit exists depends on summaries, links, and restrictions
A. Orthogonality	Proposition proved by exhibiting four systems, with second-order status supplied by stipulation	Recast as an Observation: neither property entails the other provided two possibility claims hold, namely an independently justified monitor without source discrimination and a source-discriminating detector that fails the criteria. A criterion's silence about provenance does not itself establish this; the proof is labeled an argument, and source specificity requires overlapping support at the atypicality level compared

Appendix	First version	Correction
A. Separation	Pointwise disagreement and observable separation run together; gaslight said to add states outside the original support	Cancellation made its own proposition with a two-state example; the new-support claim is stated as an assumption; scope note added that the appendix concerns report-level distinguishability, not internal organization
B. Privileged access	Budget proposition assumed a comparison system able to compute the target function without saying so until after the proof; entropy and hidden state shared a symbol; assignment probability unrestricted	Assumption moved into the statement; claim that one finite budget suffices dropped; assignment made non-degenerate; entropy written in a different symbol; remark added that informational exclusion does not establish second-order status; the deterministic proposition is said to motivate the budget framing
C. Relabeling	Said independence holds approximately when the random-label probe fails to generalize	Independence and the proxy-label distribution are to be measured on held-out data; the dependence is stated as a limitation of the control itself, making it a diagnostic whose premise can fail
C. Layer-0 probe	Stated as a proposition	Demoted to an Observation, since it is true by its antecedent; use is now a formal Definition

Appendix	First version	Correction
D. Cue design	Compared low and high cue categories; proposed rescaling a neutral state as an I = 0 condition	Binning breaks the proposition, since a pure norm detector can differ within a bin. The comparison now conditions on exact cue values or matches the whole cue distribution. Intervention now means the target steering operation, with the cue manipulation a separate factor, and low-strength steering is flagged for possibly weakening the target condition

Review suggestions declined

Not every suggestion was applied. These were left out, with the reason.

Suggestion	Reason
Name the four-part scheme as a unit, such as "the identification standard"	Adds a coined term to a corpus the author is trying to reduce
Show the final-section scheme as a table or enumerated display	The author prefers prose; the structure was clarified through topic sentences instead
Treat the arXiv identifiers and COLM venue as placeholders	They were checked against the sources and are real
Add a clause to the abstract about the limits section	The abstract already says it marks the limits of what either side has shown
Add a half-sentence in Section 8 pointing back to the dropped first limit	Judged unnecessary, since Section 4 already states it
Group the load-bearing propositions in the introduction's roadmap	Not done; the roadmap lists the four appendices by topic
Edits for a stray blank line in the bibliography	No effect on the compiled output

Suggestion	Reason
Lines from a summary of the essay, such as "tracks weirdness" and "vibes and Twitter screenshots"	Not in the essay itself. Two nearby phrasings that came close to the same overreach were softened

Still unverified

Macar et al. and Lederman and Mahowald were later checked against their full texts. The essay's descriptions held, with two wording changes: Macar's circuit is now described as early evidence-carrier features suppressing late-layer gate features in mid-depth MLP layers, and Lederman and Mahowald are now described as tentatively concluding that the models appear able to register injections, with their mixed yes-bias controls noted. The check covered all but the last few thousand characters of the Macar text.

The appendix proofs were written and revised in this conversation and checked by review comments, not by an independent mathematical reader. The orthogonality Observation is an argument from definitions, and its force depends on the second-order criteria being justified separately.

The essay has not been committed or pushed to any repository.